<table>
<tr><td>LESSON PLAN</td><td>COURSE: MHF4U</td></tr>
<tr><td>Date: Lesson #2</td><td>Unit of Study: Chapter #1 Polynomial Functions</td></tr>
<tr><td>Materials / Resources:

- Advanced Function 12 Study Guide
- Chapter #1 Formula, Aid and Homework Sheet(s)
- KWL Chart</td><td>Lesson Focus

- Transformations
- Slopes of Secants and Average Rate of Change
- Slopes of Tangents and Instantaneous Rate of Change</td></tr>
<tr><td colspan="2">Prior Knowledge:

- The roles of constants and coefficients play within a function
- Slopes of functions at specific points
- What it means to have a positive / negative slope for specific points on a function</td></tr>
<tr><td colspan="2">Overall Expectations:

1. Identify and describe some key features of polynomial functions, and make connections between the numeric, graphical, and algebraic representations of polynomial functions
2. Demonstrate an understanding o average and instantaneous rate of change, and determine, numerically and graphically and interpret the average rate of change of a function over a given interval and the instantaneous rate of change of a function at a given point</td></tr>
<tr><td colspan="2">Specific Expectations:

1.6 Determine, through investigation using technology, the roles of the parameters a, k, d, and c in functions of the form $y = af(k(x - d) + c$, and describe these roles in terms of transformations on the graphs of $f(x) = x^3$ and $f(x) = x^4$ (i.e., vertical and horizontal translations; reflections in the axes; vertical and horizontal stretches and compressions to and from the x- and y-axes)
1.1 Gather, interpret, and describe information about real-world applications of rates of change, and recognize different ways of representing rates of change (e.g., in words, numerically, graphically, algebraically)
1.2 Recognize that the rate of change for a function is a comparison of changes in the dependent variable to changes in the independent variable, and distinguish situations in which the rate of change is zero, constant, or changing by examining applications, including those arising from real-world situations (e.g., rate of change of the area of a circle as the radius increases, inflation rates, the rising trend in graduation rates among Aboriginal youth, speed of a cruising aircraft, speed of a cyclist climbing a hill, infection rates)
1.3 Sketch a graph that represents a relationship involving rate of change, as described in words, and verify with technology (e.g., motion sensor) when possible</td></tr>
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1.4 Calculate and interpret average rates of change of functions (e.g., linear, quadratic, exponential, sinusoidal) arising from real-world applications (e.g., in the natural, physical, and social sciences), given various representations of the functions (e.g., tables of values, graphs, equations)
1.5 Recognize examples of instantaneous rates of change arising from real-world situations, and make connections between instantaneous rates of change and average rates of change (e.g., an average rate of change can be used to approximate an instantaneous rate of change)
1.6 Determine, through investigation using various representations of relationships (e.g., tables of values, graphs, equations), approximate instantaneous rates of change arising from real-world applications (e.g., in the natural, physical, and social sciences) by using average rates of change and reducing the interval over which the average rate of change is determined
1.7 Make connections, through investigation, between the slope of a secant on the graph of a function (e.g., quadratic, exponential, sinusoidal) and the average rate of change of the function over an interval, and between the slope of the tangent to a point on the graph of a function and the instantaneous rate of change of the function at that point
1.8 Determine, through investigation using a variety of tools and strategies (e.g., using a table of values to calculate slopes of secants or graphing secants and measuring their slopes with technology), the approximate slope of the tangent to a given point on the graph of a function (e.g., quadratic, exponential, sinusoidal) by using the slopes of secants through the given point (e.g., investigating the slopes of secants that approach the tangent at that point more and more closely), and make connections to average and instantaneous rates of change
1.9 Solve problems involving average and instantaneous rates of change, including problems arising from real-world applications, by using numerical and graphical methods (e.g., by using graphing technology to graph a tangent and measure its slope)

Learning Goals:

- Students will be able to determine the different transformations that are applied to a base function
- Students will understand the connection between average rate of change and the slope of a secant, and instantaneous rate of change and the slope of the tangent
- Students will learn to apply numerical and graphical methods to calculate and interpret average and instantaneous rate of change in real-world applications that involve polynomial functions

Success Criteria:

1. By the end of this lesson I will be able to describe the transformation associated with the roles of the parameters a, k, c, and d, in polynomial functions of the form $y = a[k(x - d)]^n + c$
2. By the end of this lesson I will be able to differentiate between slope of secant and slope of tangent, and how they are average rates of change and instantaneous rates of change, respectively
3. By the end of this lesson I will be able to find average and instantaneous rate of change by using numerical and graphical methods

Assessment Strategies:

Formative: homework for chapters 1.4, 1.5 & 1.6

Summative: test #1

Time: **Part 1**	**Introduction:** Take up / check homework from lesson #1 and check for student’s understanding. If needed, go over specific questions.
Time: **Part 2**	**Lesson:** <u>Chapter 1.4 Transformation</u> Explain what it means for a function to be transformed. Go over each individual parameter to explain what each stands for (e.g., for function $y = a[k(x - d)]^n + c$, a = vertical stretch / compression, k = horizontal stretch / compression, d = horizontal translation, c = vertical translation), and what it means for a & k to be less or greater than 1. Ensure the steps to applying transformation. First, make sure the function is in the form $y = a[k(x - d)]^n + c$. Second, apply any horizontal / vertical stretches or compressions. Third, apply any reflections. Fourth, apply any horizontal or vertical translations. <u>Chapter 1.5 Slopes of Secants and Average Rates of Change</u> Define slopes of secants and rates and how it is related to average rates of change. Identify that a rate of change is a measure of how quickly one quantity changes with respect to another. Make connections between slopes of secants with real-world applications. <u>Chapter 1.6 Slopes of Tangents and Instantaneous Rates of Change</u> Define slopes of tangents and how it is related to instantaneous rates of change. Identify the similarity and different between slopes of tangents and slopes of secants. Compare and contrast average rates of change and instantaneous rates of change. Make connections between slopes of tangents with real-world applications.
Time: **Part 3**	**Wrap-up:** Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions. Use KWL chart to better understand student’s learning needs for chapter #1.

Notes, reminders & homework:

<u>Reminder</u>:

- Quiz #1 (to be written before / after current lesson)
- Test #1 (to be written before / after lesson #5)

<u>Homework</u>:
Chapter 1.4: #1 – 6, 7 a), 8 – 10, 12
Chapter 1.5: #1 – 7, 11 – 15
Chapter 1.6: #1 – 10, 13

Chapter #1: Challenge Questions (preparation for test #1)
C1 – C4, C9

Lesson Reflection: